



Disentangled representational learning for anomaly detection in single-lead electrocardiogram signals using variational autoencoder

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ABSTRACT

Wearable technology enables the unsupervised recording of electrocardiogram (ECG) signals. Analyzing these high-dimensional ECG data poses challenges regarding statistical approaches and explainability. This work investigates the feasibility of medically explainable anomaly detection through disentangled representational learning of ECGs and personalization to mitigate inter-subject variations. Five open-source ECG datasets were converted into a set of denoised one-second traces of lead I signal, each covering individual features such as wave morphologies and pathologies. A beta total correlation variational autoencoder was optimized on four of these datasets for 68 systematic parameterization variants. The best-performing model revealed disentanglement in the 12-dimensional embedding space, specifically between atrial- and ventricular features. Within the embedding space, a k-nearest neighbor classifier was evaluated on a left-out test set tailored for anomaly detection. The result is a F1 score of 0.94 for the binary prediction of sinus rhythm and the pathological classes: Left bundle branch block, right bundle branch block, myocardial infarction, and AV block (1st degree). The 90.94% accuracy in anomaly detection falls within the range of established detectors (89.38%–99.77%) but offers the advantage of being explainable and largely unsupervised. Model fine-tuning for each of 100 randomly sampled individuals of the Icentia11k dataset mitigated inter-subject variations. The associated F1 score for predicting normal, premature atrial contraction, and premature ventricular contraction from the embedding space was 0.93. The distribution plots of pathologies along the explainable axis were reasonably consistent with medical expertise. The results suggest the presented disentangled variational autoencoder as a robust method for explainable ECG representation.

1. Introduction

Electrocardiography provides a noninvasive approach for the assessment of cardiovascular health and diagnosing potential anomalies. Regular screening for heart conditions is a prerequisite for the prevention of long-term diseases and early intervention. According to estimates, half a billion people worldwide were affected by cardiovascular disease (CVD) in 2021, resulting in 20.5 million deaths [1]. The United Nations is actively working towards reducing CVDs as the leading cause of premature death as part of its sustainable development goals [2].

Wearable technology contributes to this objective by facilitating the recording of electrocardiogram (ECG) signals independent of time, location, and clinical environment. The quantified self movement helps to address healthcare challenges by enabling continuous monitoring,

reducing the burden on clinical infrastructure, and generating a wealth of research data. Further, the shortage of healthcare professionals and the time-consuming nature of manual ECG analysis require a shift towards decentralized and automated analysis. Computational decision-making offers promising perspectives, such as objective and reproducible results, personalization approaches, and seamless and unobtrusive anomaly detection. However, challenges remain in unsupervised environments, particularly concerning the lack of contextual information and limited interpretation.

Representational learning is an approach to machine learning in unsupervised settings that aims to extract latent and generative factors from high-dimensional data. In the context of ECG signals, the latent factors might describe the main morphological characteristics, e.g., the P-wave, the QRS-complex, and the T-wave (cf. Fig. 1). Mapping

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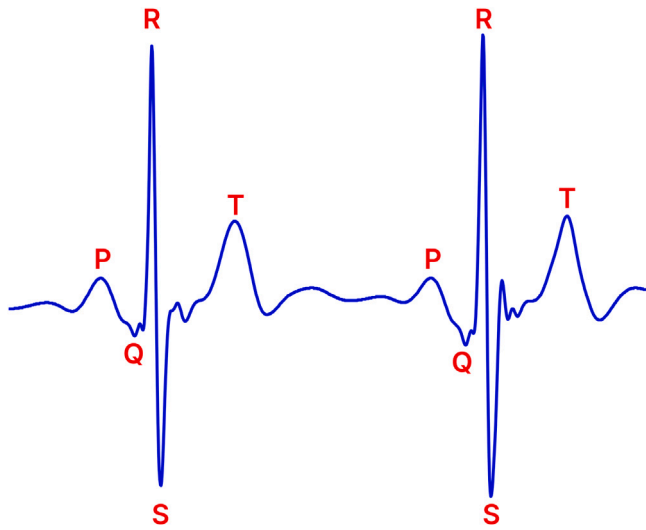


Fig. 1. Main morphological landmarks in the lead I ECG.

data from a high-dimensional space into a latent vector space, i.e., a low-dimensional representation, enables the application of anomaly detection methods, such as visual inspection, classical statistics, and clustering, without suffering from the curse of dimensionality [3].

Autoencoder (AE) is a method of representational learning that incorporates deep learning to operate on complex data such as images, time series, and graphs [4]. The approach aims to encode latent factors from usually high-dimensional data and reconstruct (decode) the input data from these embeddings. AE applications and extensions are diverse, ranging from signal denoising and data compression to anomaly detection across almost all domains [5,6]. Recent efforts suggested the application of AEs to the cardiology domain to demonstrate its feasibility [7]. In particular, Variational Autoencoders (VAEs) were used for several investigations due to the inclusion of a probabilistic component that ensures specific properties of the embedding space and allows the sampling of synthetic data.

VAEs have demonstrated the ability to realistically generate single-lead ECGs that represent an isolated cardiac cycle, accomplished through convolutional neural networks for encoding and decoding processes [8]. 25 latent factors were extracted from preprocessed ECG signals and used to generate synthetic ECG samples. The visual outputs indicate that these latent factors can accurately capture the underlying signal characteristics. The maximum mean discrepancy metric, which measures the proximity to real-world ECG, yielded a value of 3.83×10^{-3} , further confirming the precision of the generated signals. Some latent factors also proved clinically interpretable and provided insights into specific ECG factors, such as the height of the T-wave or the presence of ST-depression.

In the study by Jang et al. (2021), Convolutional Variational Autoencoders (CVAEs) were investigated using a dataset of 596 000 ECG samples and three validation datasets [9]. Manual explorations and visualizations of the latent vector space revealed a strong correlation with the enclosed arrhythmia labels, e.g., sinus bradycardia, supraventricular tachycardia, and atrial flutter. Employing a tree-based model applied to the encoded ECGs yielded an F1 score of 0.86 for predicting the arrhythmia classes. In closely related work, stacked convolutional autoencoders were optimized using the MIT-BIH dataset and achieved an accuracy of 98.96% and a sensitivity of 97.32% [10,11]. The findings of this research highlight that CVAEs can uncover distinctive characteristics present in various types of ECGs. Furthermore, the study suggests that CVAEs offer a viable alternative to traditional manual feature extraction methods.

Researchers have conducted a range of similar studies, including the detection of hypoglycemic events in the latent factors of ECGs [12,13],

Atrial Fibrillation classification [14], signal denoising with subsequent classification [15], or application of VAE in the case of imbalanced datasets [16]. The latter leveraged BI-LSTM layers and experiments showed an F1 score of at least 0.90 for the task of ECG anomaly classification.

Whereas the initial efforts showed promising results, representational learning requires a more thorough understanding of the latent dimensions and factors. Specifically, this involves interpreting the dimensions of the latent vector space as a representation of ECG morphologies and, consequently, explaining anomalies that occur as linear combinations of latent factors. It should be noted that the term factor disentanglement is synonymous with interpretable, as a clear association between the factors of the data and the latent dimensions implies disentanglement by definition.

Li et al. (2022) introduced a time series embedding framework to achieve disentanglement of the latent factors at multiple levels [17]. Therefore, the VAE loss term was decomposed to find the evidence linking the factorial representations. Furthermore, a regularization term was added to maximize the mutual information between the observation and latent spaces to address the challenge of vanishing gradients in VAE architectures. The method was validated using an ECG dataset with diverse T-wave patterns and associated QT-intervals [18]. The disentangled encoding of the ECG into 8 dimensions revealed interpretable factors, some strongly associated with ECG wave morphologies.

SimECG is a synthetic 12-lead ECG dataset generated from known inter-subject anatomical factors, e.g., cardiac rotation, and a factor representing the location of ventricular activation [19]. Gayawali et al. (2021) investigated how the relaxation of the standard β -VAE regularization improved model capacity and disentangled the generative factors in a controlled environment. Specifically, the relaxation allowed for an independent latent factor model where a stochastic Beta-Bernoulli Process models the latent variable [20]. The method demonstrated a strong disentanglement along the generative factors and, at the same time, significantly improved the clinical task of detecting the origin of ventricular activation. The results and interpretations highlighted the importance of considering inter-subject variations in automated analysis and motivated further exploration of disentanglement in VAEs. Prior to this, Gayawali et al. (2018) employed a conditional VAE for the task of localizing the origin of ventricular tachycardia (VT) in 12-lead ECG [21]. The maximum mean discrepancy measured the independent representation of task and inter-subject factors, whereas weakly supervised contrastive regularization was used to learn the task-specific representation. The results indicate improved performance in localizing VTs for the proposed method, accounting for inter-subject variation.

A simple VAE architecture with a $\beta > 1$ weighted Kullback–Leibler divergence and a latent dimension of 10 is introduced in Steenkiste et al. (2019) to detect beat anomalies in the MIT-BIH Arrhythmia dataset [11,22]. Compared to a standard AE, the proposed technique yielded high classification accuracy and disentangled the latent factors, thus providing explainability. Follow-up work by Leur et al. (2022) further emphasized the necessity for the investigation of explainability in future research [23]. This emphasis arises from the aim to mitigate model bias, enhance clinical relevance, and instill greater confidence in the application of artificial intelligence. Moreover, the article introduced a framework known as FactorECG, which is an explainable VAE, leveraging the inherent generative nature of β -VAEs, in which the β parameter enforces disentanglement of the latent factors. Thereby, the architecture of FactorECG was inspired by a more generic VAE for time series data [24]. The interpretation of the latent dimensions of FactorECG was evaluated by decoding several ECGs from 21 latent factors by changing the values along individual embedding axes and visually examining the changes in the generated signals. Validation on an external dataset produced results comparable to those of black box models. The mean area under the receiver operating curve (AUROC) was 0.94 for conventional ECG classification, 0.90 for detecting reduced

ejection fraction, and 0.76 for predicting 1-year mortality. The results were also compared to MUSE (GE HealthCare), a traditional rule-based anomaly detection system used in clinical practice and aligned with the SCP-ECG standard coding convention [25]. The comparison revealed that the traditional system exhibited reduced performance, with a mean AUROC of 0.73. These outcomes underscore the potential of the FactorECG framework. To further improve the explainability of the model in a task-specific context, Valk et al. (2023) proposed to jointly optimize the β -VAE and a task-specific loss, such as the functionality of the left ventricle [26]. Similar to the visualization technique of FactorECG, an explainable approach is ensured but further optimized by keeping the size of the latent vector space small and thus more accessible. The approach showed improved prediction accuracy, reconstruction performance, and explainability compared to the standard VAE. A partially similar yet more comprehensive framework, which necessitates annotated data for explainable machine learning on multivariate time series, is discussed in Todo et al. (2023) [27]. The approach involves a contrastive VAE that regularizes the latent space into general and class-based features, resulting in a well-structured latent space. Experimental results on 12-lead ECG data demonstrated precise reconstruction and revealed distinct factor-based cluster behavior within the latent space.

Machine learning on health data without professional supervision requires comprehensive technological considerations to ensure high standards of traceability and reliability. This is especially important as medical conclusions can significantly impact an individual's well-being. In this context, the rapidly evolving landscape of machine learning research reveals gaps in the current state of explainable and unsupervised ECG anomaly detection. For example, employing modifications of β -VAE to achieve stronger disentanglement and more accurate reconstructions, particularly for rare diseases, remains an open topic for research, as proposed by Leur et al. (2022) [23]. Furthermore, related work is largely tailored to clinical ECGs and is often constrained to a single database in evaluation, which limits generalizability and raises the risk of inducing bias [28]. Additionally, these studies rarely consider inter-subject variations induced in heterogeneous environments — an aspect that is becoming increasingly important due to the prevalence of wearable devices. Moreover, the focus tends to be on detecting a predefined set of anomalies rather than enabling a more generalized approach to outlier detection. In this context, Steenkiste et al. proposed verifying disentangling methods with multiple beat annotations beyond the MIT-BIH dataset as future work [22].

The challenges and gaps identified in existing methods motivated this work. The need for a robust framework that can detect ECG anomalies in an unsupervised manner, while also accounting for rare diseases and inter-subject variations, is increasingly relevant, especially as wearable devices capable of recording ECGs become daily companions. Current models are often limited by their reliance on single datasets and a narrow focus on predefined anomalies, which restricts their applicability and generalizability across diverse populations. To address these shortcomings, this work investigates recent advances in disentangled representational learning and employs an atomic approach to capture single ECG morphologies. Specifically, this work pursues a rigorous and explainable approach by detecting outliers as combinations of anomalies in atomic features. For instance, a myocardial infarction may manifest as a combination of the atomic features that indicate an altered ST-segment, a pathological Q-wave, and a loss of R-wave amplitude. By further utilizing multiple diverse datasets, the proposed method aims to model a broader range of ECG variations, including rare conditions, and improve the detection of outliers in heterogeneous environments. Ultimately, this approach not only enhances model explainability but also has the potential to revolutionize cardiac care by enabling accurate, real-time detection of cardiac anomalies in real-world environments, including continuous monitoring through wearable devices.

The general focus of this article was on the explainable representational learning of single morphological factors of multi-source ECG

Table 1
Data sources and properties.

Dataset	Samples	Subjects	Frequency	Leads	Length	Recording device
PTB-XL	21 837	18 885	400 Hz	12	10 s	Schiller
CU-SPH	10 646	10 646	500 Hz	12	10 s	Zhejiang
lcentia11k	11 000	11 000	250 Hz	1	58 h	CardioSTAT
MedalCare-XL	16 900	—	500 Hz	12	10 s	Synthetic
ECGSYN	250	—	500 Hz	1	50 s	Synthetic

signals, such as the shape of the P- and T-wave, the QT-interval, and the width of the R-peak, rather than heart pathologies. The application area is limited to single-lead ECGs, as commercially available wearables can record these in a mobile environment. β -Total Correlation Variational Autoencoders (β -TCVAEs) are employed as a method for disentangling single morphological factors along the dimensions of the latent vector space [29], offering a refinement over the β -VAE approach. The following three research questions are the subject of the investigation:

- 1. Q1: Disentangled VAEs for ECG Representation**
To what extent can β -TCVAEs effectively represent the morphological factors (e.g., P-wave, QRS-complex, T-wave) of single-lead ECG in a low-dimensional vector in a disentangled fashion?
- 2. Q2: Anomaly Explanation and Generalizability**
How do disentangled representations derived from β -TCVAEs enhance the explainability of anomalies in ECG signals? Specifically, can the investigation of the latent dimensions in β -TCVAEs explain and identify abnormal ECG factors related to specific cardiovascular conditions? How does anomaly detection perform specifically when applied to previously unseen subjects and data?
- 3. Q3: Mitigating inter-subject variations**
Can personalized VAE models, tailored to individual patient data, improve anomaly detection accuracy in ECG signals by effectively mitigating inter-subject variability?

The article presents material and methods, including data sources (2.1), preprocessing to ensure homogeneity of data from different sources (2.2), theory and implementation of VAE disentanglement (2.3 and 2.4), and evaluation (2.5). The results (3), categorized according to the proposed research questions (Q1–Q3), are then discussed to address feasibility, limitations, and future research directions (4).

2. Materials and methods

2.1. Data

The material used in this study is open-source data. Three sources of real-world ECG data [30–32] and two synthetically generated datasets [33,34] formed the basis for machine learning routines and evaluation (Table 1). The main inclusion criterion was the availability of Einthoven's lead I at a large scale and metadata such as beat annotations and signal characteristics.

The Physikalisch Technische Bundesanstalt (PTB) released PTB-XL in the context of a long-term project for data curation and research for novel algorithmic approaches [30,35,36]. A total of 18 885 distinct patients with annotated demographic information such as gender and age were included alongside a total number of 21 837 ten-second 12-lead ECGs at a sampling rate of 400 Hz. The dataset annotation conforms with the SCP-ECG standard [25], containing 71 unique labels. The ratio of ECG traces showing abnormalities is 17% and is close to the general prevalence of 15 % for CVDs [37]. Thus, PTB-XL applies for an appropriate candidate to represent the distribution of CVDs among the global population.

Chapman University (CU) and Shaoxing People's Hospital (SPH) published a 12-lead ECG database for arrhythmia research covering

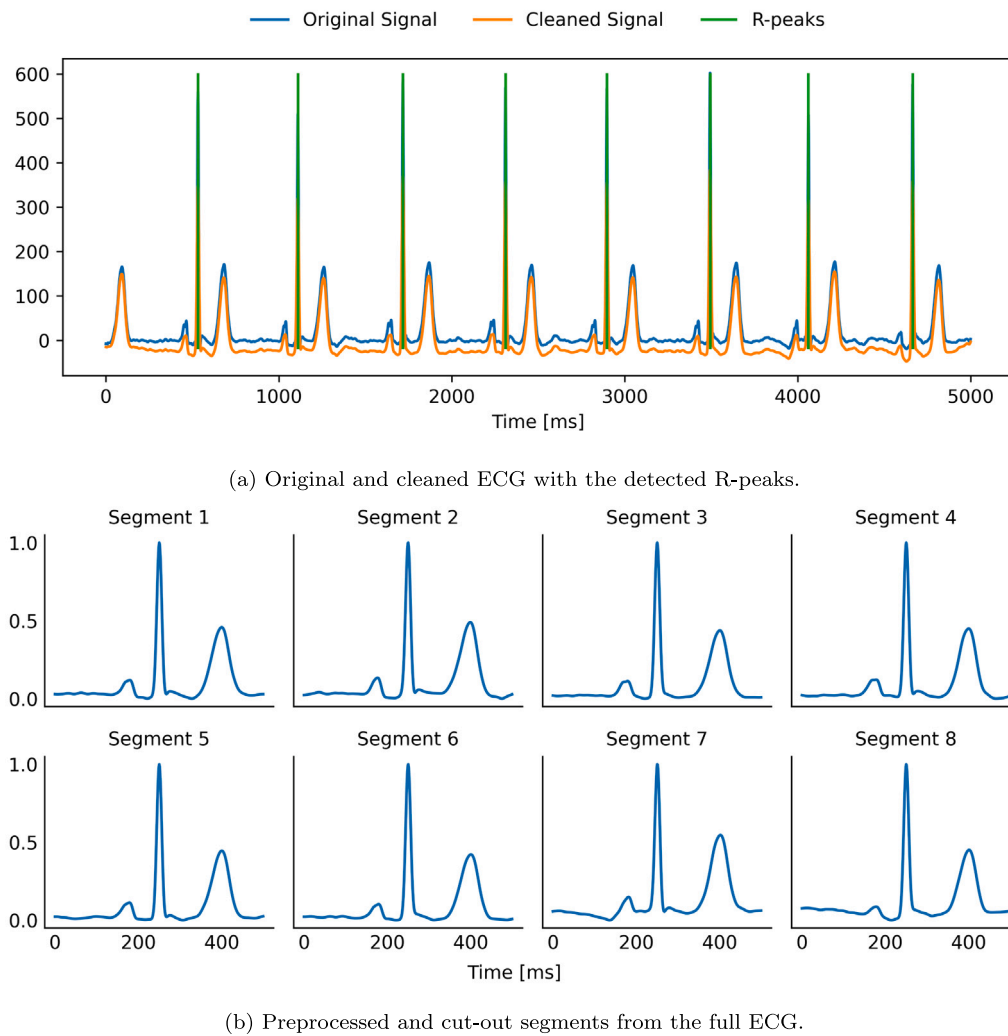


Fig. 2. Preprocessing pipeline using a sample of CU-SPH.

10646 subjects [31], denoted as *CU-SPH*. The ECGs show a length of 10 seconds at a sampling rate of 500 Hz, and each was assigned to one of 11 rhythm schemes, such as atrial fibrillation, sinus bradycardia, or the normal sinus rhythm. The dataset also details beat anomalies (67 unique classes) and information on ECG morphologies, e.g., QT-interval, T-offset, and QRS-count. A certified physician conducted the annotation process, and a second physician performed a validation round. In the case of mismatched labeling, a third senior physician assessed the ECG. The percentage of healthy-rated ECG is 17 %, i.e., most signals show any sign of an anomaly.

Algorithms engineering in the field of representational learning of electrocardiogram signals motivated the release of *Icentia11k* [32,36,38]. The dataset comprises 11000 unique patients, each composed of 50 segments of approximately 70 min, resulting in a total duration of ~ 58 h at a sampling rate of 250 Hz. Information about annotations is available on the beat level, specifically at the R-peak's locations. The categories for beat classification are either premature atrial contraction (PAC), premature ventricular contraction (PVC), standard, undefined, or aberrated. Additional data include demographic details (gender, age) and a classification of rhythm (normal sinus rhythm, atrial fibrillation, atrial flutter, noise, undefined). The authors report no preprocessing steps except segmenting and rearranging the raw signals into smaller chunks.

Simulating the heart's electrical activity produced a total of 16900 synthetic 12-lead ECG [33]. The primary motivation for synthetic sampling was the lack of clinical data, e.g., for underrepresented

pathologies, and employing the database while validating machine-learning-based ECG tools. The *MedalCare-XL* database divides the ECGs into eight classes: sinus; right bundle branch block (RBBB); left bundle branch block (LBBB); myocardial infarction (MI); 1st degree AV block (AV); left atrial enlargement (LAE); interatrial conduction block (ICB); fibrotic atrial cardiomyopathy (FAC). MI was further divided into the subclasses right-anterior descending (RAD), left-anterior descending (LAD), and left circumflex (LCX). The dataset is already split into training (71.38%), validation (14.45%), and test (14.17%) segments, enabling objective, reproducible, and comparable results, particularly for anomaly detection.

ECGSYN is a realistic waveform generator, parameterized to produce various morphologies [34,36]. The synthetic generation of ECG data allows for the systematic validation of model disentanglement, leveraging the control over generative factors in a simulated environment. 250 synthetic single-lead ECGs are produced, each averaging 50 heartbeats with a standard deviation of 2, incorporating additive Gaussian noise. The generation process involves modifying parameters such as height, angle, and width for the P-, Q-, R-, S-, and T-wave (15 features in total). Altering multiple parameters revealed distorted and unrealistic signals, except when simultaneously altering P- and T-wave heights. Thus, all values were kept constant during signal generation, except one varying factor. Each factor had ten gradations, i.e., 15×10 ECGs were generated in the first step. The combinations of P- and T-wave height for ten gradations each resulted in a further 10×10 ECGs.

2.2. Preprocessing

The preprocessing step was designed to ensure that the samples have consistent and uniform statistical characteristics without altering the relevant morphologies. The objective is to produce denoised ECG traces of one-second length sampled at 500 Hz with the R-peak forming the center (cf. Fig. 2). This lowers the degree of freedom of the data and simplifies the required complexity of the machine learning model.

The cleaning process involves extracting the lead I information from the raw data and then applying a fifth-order Butterworth filter, followed by a powerline filter. The R-peaks are then detected (cf. Fig. 2(a)) using the neurokit peak detection method [39]. Subsequently, the signal was resampled to a frequency of 500 samples per second using spline interpolation. Data normalization was performed to map the values into the zero–one range after isolating approximately single heartbeats by cutting 250 points to each side of the R-peaks (cf. Fig. 2(b)). Though the information on absolute electrical deflection in millivolts is a relevant diagnostic parameter for some pathologies, normalization further reduces the complexity that the statistical model has to intercept.

All available metadata was assigned to the samples for later analysis purposes. For datasets like *PTB-XL*, *CU-SPH*, and *MedalCare-XL*, where metadata is provided for the entire signal rather than on a beat-by-beat basis, the same label was applied to all coherent samples.

A stand-alone repository¹ contains the code for loading and preprocessing the signals from the respective data sources as TensorFlow Datasets [40,41]. The structure enables transparency, reproducible results, and convenient integration of the data sources in consecutive research.

2.3. Disentangled variational autoencoder

A latent representation, denoted as $z \in V \subset \mathbb{R}^k$, seeks to describe a data sample $x \in X \subset \mathbb{R}^n$ with less information, that is $k \ll n$, while preserving relevant features such as similarities among data samples in X . The theoretical feasibility of an effective latent representation was explored using the example of the Johnson–Lindenstrauss lemma [42] and practically investigated with techniques such as principal component analysis [43], t-distributed stochastic neighborhood (t-SNE) [44] and VAEs.

Autoencoders consist of an encoder $E_\phi : X \rightarrow V$ and a decoder $D_\theta : V \rightarrow X$, both neural networks parameterized by weights ϕ and θ , respectively. The encoder maps high-dimensional data $X \subset \mathbb{R}^n$ to a latent vector space $V \subset \mathbb{R}^k$. In contrast, the decoder aims to reconstruct the original signal from this compressed representation. The primary objective is to minimize a reconstruction error $\mathcal{L} : X \times X \rightarrow \mathbb{R}$ by iteratively adjusting the weights θ and ϕ . Formally, the optimization objective reads

$$\min_{\theta, \phi} \sum_{x \in X} \mathcal{L}(x, (E_\phi \circ D_\theta)(x)). \quad (1)$$

The objective function in Eq. (1) facilitates encoding the k most significant latent factors of the signal, as otherwise, the decoder lacks the capability for high-quality reconstruction.

Variational Autoencoders (VAEs) provide a probabilistic perspective and introduce regularization on the latent representation to address the continuity and completeness of the embedding space. A distribution parameterized by θ assigns a likelihood to each data sample $x \in X$ denoted as $p_\theta(x)$. Modeling the relationship between the data sample and the latent representation involves marginalizing over $z \in V$.

$$\begin{aligned} p_\theta(x) &= \int_z p_\theta(x, z) dz \\ &= \int_z p_\theta(x|z) p_\theta(z) dz \\ &= \int_z p_\theta(z|x) p_\theta(x) dz \end{aligned} \quad (2)$$

The terms $p_\theta(x|z)$, $p_\theta(z|x)$, and $p_\theta(z)$ in Eq. (2) are generally referred to as likelihood, posterior, and prior. The objective is twofold and includes, on the one hand, finding the parameterization θ that maximizes $p_\theta(x)$ for the occurrence of observed data samples $x \in X \subset \mathbb{R}^n$. On the other hand, the calculation of the posterior $p_\theta(z|x)$ is crucial for investigating the latent representation but remains challenging to compute due to the intractable integral involved in marginalization over z . In the context of the autoencoder architecture, the posterior is associated with the encoder (inference network), and the likelihood is associated with the decoder (generative network). Using the inference network as an approximation of the posterior ($E_\phi = q_\phi(z|x) \approx p_\theta(z|x)$) leads to the objective of minimizing the Kullback–Leibler (KL) divergence $D_{KL}(q_\phi(z|x) \parallel p_\theta(z|x))$ w.r.t. ϕ . An equivalent formulation of the problem can be obtained through calculus and rearrangement of the KL term. Formally,

$$\begin{aligned} \log p_\theta(x) - D_{KL}(q_\phi(z|x) \parallel p_\theta(z|x)) \\ = -\mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] \\ + \beta D_{KL}(q_\phi(z|x) \parallel p_\theta(z)) \end{aligned} \quad (3)$$

in which $p_\theta(x)$ is approximated by the generator network for $\beta = 1$. The first term of the left-hand side of Eq. (3) describes the reconstruction capabilities, whereas the second term ensures adherence to the regularization of the latent vector space. The right-hand side of Eq. (3) represents a lower bound of $\log p_\theta(x)$ and is denoted as the evidence lower bound (ELBO).

In β -VAE [45], the coefficient β regulates the strength with which encoded samples adhere to the prior distribution, influencing disentanglement. A higher β has shown stronger disentanglement but may lead to decreased reconstruction accuracy. Further decomposing the Kullback–Leibler term reveals three distinct components, as shown in Chen et al. (2018), shedding light on the sources of disentanglement, particularly the β total correlation term [29]:

$$\begin{aligned} \mathcal{L}_{\theta, \phi}(x) &= \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] \\ &\quad - \alpha I_q(z; x) \\ &\quad - \beta D_{KL}(q_\phi(z) \parallel \prod_j q(z_j)) \\ &\quad - \gamma \sum_j D_{KL}(q_\phi(z_j) \parallel p(z_j)) \end{aligned} \quad (4)$$

The parameter α regulates the mutual information function I , β the total correlation term, and γ the dimensionwise Kullback–Leibler divergence. The evaluation of the distribution $q_\phi(z)$ in Eq. (4) requires the calculation of the entire dataset in each pass of a batch, which can lead to a massive increase in computational time and resources. In Esmaili et al. (2019), an approximation of $q_\phi(z)$ was presented, which was used as a consistent estimator, where N is the length of the complete dataset and B is the batch size [46]:

$$\hat{q}_\phi(z) = \frac{1}{N} q_\phi(z|x) + \frac{N-1}{N(B-1)} \sum_{j \neq j'} q_\phi(z^j | x^{j'}) \quad (5)$$

A batch size of 512 to 1024 is recommended for a stable estimate and likely low bias of Eq. (5) [46].

2.4. Implementation

In the broadest sense, the architecture of the underlying model is inspired by the architecture of FactorECG [23]. The encoder structure consists of four convolutional blocks with doubling filter size and leaky rectified linear unit (ReLU) activation, followed by a max-pooling layer and two dense mapping layers. The decoder is a mirrored version of the encoder that maps the latent vector into the shape (500,) using two consecutive dense layers followed by four transposed convolutional layers with leaky ReLU activation (cf. Fig. 3). Convolutional layers are frequently used in the field of ECG classification due to their strong performance and feature extraction capabilities [28,47], making them

¹ <https://github.com/CardioKit/ECG-TFDS>. Retrieved on: 23 October 2024.

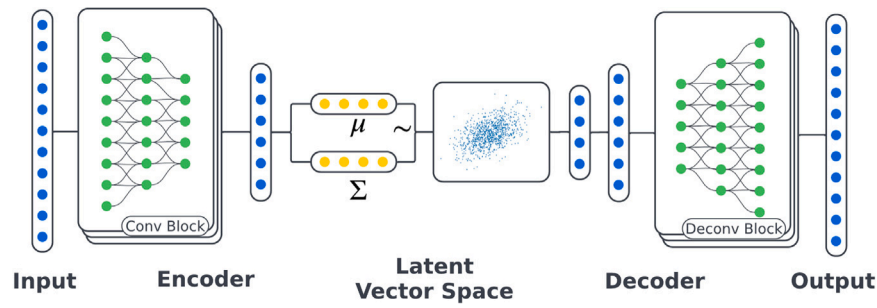


Fig. 3. Model architecture.

well-suited for identifying abstract representations of the data. Max-pooling layers, which propagate the maximum value from one layer to the next, are employed to reduce dimensionality while emphasizing and propagating the most important features [48]. This is particularly valuable because max-pooling provides translation invariance [48], ensuring robustness even when timeseries features shift slightly. Finally, the leaky ReLU activation function is used due to its proven ability to mitigate the vanishing gradient problem and effectiveness in dealing with non-activated units during optimization [49,50]. The selection of a multivariate Gaussian distribution as the prior is motivated by the premise that ECG parameters conform to a normal distribution, akin to numerous other health parameters such as body height. The parameters of the corresponding Gaussian distribution were set by the mean vector $\mu = 0$ and covariance matrix $\Sigma = I$.

Variational autoencoders can face optimization issues such as mode collapse and posterior collapse. The latter describes the phenomena of a posterior that strongly approximates the prior distribution without considering the underlying structure of the data. Mode collapse is the inability to reconstruct a diverse range of data but collapses to a static output. Both phenomena are rooted in overregulation of the divergence term [51] and require finding a tradeoff towards the reconstruction loss. Accordingly, the weight terms α , β , and γ in Eq. (4) need to be set properly. A coarse-to-fine hyperparameter search was performed on the dimension of the latent vector space and the coefficient triple (α, β, γ) to find a suitable tradeoff. Furthermore, an algorithm, not yet rigorously proven, to disable the optimization of the decoder in the case of a lagging inference network is implemented as a callback function after each epoch to prevent posterior collapse [52]. The grid search values for the latent dimension were set to 8, 12, 16, 20, and 24, in line with related work [8,17,22,23]. The initial systematic search for the regularization term weights was broad, with results such as reconstruction quality and vector space encodings thoroughly analyzed. In the second search, the value range was narrowed to refine the search around the best parameter combination identified in the first round. To further mitigate the effect of mode collapse, a KL-annealing strategy was employed, where the values of (α, β, γ) were gradually increased over each epoch. However, this approach had only a minimal effect.

Except for the *Icentia11k* dataset and *Medalcare-XL*'s test dataset, all introduced datasets from Section 2.1 were included in the optimization routine of the model using the RMSprop algorithm [53]. Each experiment ran for 200 epochs with an early termination of 50 rounds on the *MedalCare-XL* validation dataset loss and a batch size of 1024 samples. The start learning rate was set to 0.002, whose value was halved for a non-decreasing reconstruction error throughout 20 epochs. The numbers were chosen from previous experiments that provided stable optimization for the particular configuration.

Inter-subject variations and heterogeneous environments can result in an individual fingerprint in the encoding distribution. The result is a clustering concerning individuals rather than the relevant factors. Therefore, the model was fit to each of 100 randomly selected subjects of the *Icentia11k* dataset for another 20 epochs. The fine-tuned model makes it possible to evaluate the potential of personalized

models, which is particularly interesting for decentralized approaches, e.g., cross-device federated learning for mobile environments.

The code was run on a Linux-based operating system with an *NVIDIA GeForce RTX 4090* for accelerated computing. The software is available in a dedicated repository.²

2.5. Evaluation

During the optimization and evaluation of the model using different combinations of hyperparameters, key measurements were taken, and various results were recorded, including: (1) the mean absolute error (MAE) to assess goodness of fit and quality of reconstruction; (2) the weights and losses of the regularization terms (mutual information, total correlation, dimension-wise KL); (3) plots of the reconstruction; (4) the model with the best performance in terms of loss; (5) the mutual information gap (MIG) of the ECGSYN embedding to measure disentanglement. The MIG is defined as

$$\frac{1}{K} \sum_{k=1}^K \frac{1}{H(v_k)} \left(I_n(z_{j^{(k)}}; v_k) - \max_{j \neq j^{(k)}} I_n(z_j; v_k) \right), \quad (6)$$

where $j^{(k)} = \arg\max_j I_n(z_j; v_k)$ for an embedding $z \in \mathbb{R}^n$ and factors $v \in \mathbb{R}^K$ [29]. The functions H and I represent the Shannon Entropy and mutual information, respectively. Given that certain features exhibit interdependencies, such as the T-wave morphology and QT-interval, the MIG was determined by focusing on two distinctly different attributes: the P-wave and T-wave height within the ECGSYN dataset, as these morphologies are associated with atrial and ventricular contraction, two independent cardiac activities. By definition, the MIG helps to identify models of disentanglement capabilities and thus contributes to the objective of explainability. Finally, all experiments and their metrics were compiled into the Appendix for comparison. The model that performed best in MIG and showed decent performance of loss metrics was selected for further evaluation, strictly following the scientific questions outlined in the introduction. Table 2 presents a detailed overview of the key steps involved in model training and evaluation, highlighting the specific datasets used at each stage. This includes the evaluation of disentanglement, anomaly detection, and personalization.

Disentanglement

The evaluation of the model's ability to incorporate distinct factors into its dimensions considered how well the model's embedding space aligned with the generative factors of ECG waveforms. This analysis was focused explicitly on embeddings from the *MedalCare-XL*, *CUSPH*, and *ECGSYN* datasets, which offer access to single generative ECG factors like T-wave height, P-wave width, and QT-interval. A linear regression approach was used to estimate these generative factors from the embedding dimensions. The predictability of these dimensions and the information they provided about the factors were measured using the coefficient of determination (R^2). The factor that showed the

² <https://github.com/CardioKit/VECG>. Retrieved on: 23 October 2024.

Table 2
Overview: Data sources and their integration into the research.

Dataset/Activity	PTB-XL	ECGSYN	CU-SPH	MedalCare-XL		Icentia11k
				Train	Validation	Test
Model Training	✓	✓	✓	✓		
Model Validation					✓	
Disentanglement		✓	✓			
Anomaly Training				✓		
Anomaly Validation					✓	
Anomaly Testing						✓
Personalization						✓

highest R^2 for a dimension was set as the computational interpretation of the dimension.

A scatter plot was created for the two dimensions with the highest R^2 , showcasing (1) the embedding's feature-specific orthogonal structure and (2) the distribution of characteristics across a dimension. The analysis also included an iterative method where all embedded values were fixed, and one dimension was systematically adjusted to observe changes in the reconstruction, a technique close to that used in FactorECG. The description of the plots was set to the most likely axis interpretations given by the highest R^2 values. In addition, the evaluation of the observed changes by an expert from the field of cardiology was included.

Anomaly detection

The *MedalCare-XL* dataset offers a comprehensive collection of pathology annotations, making it ideal for testing anomaly detection within embedded and explainable frameworks. It is important to note that not every annotated anomaly can be detected within a single heartbeat from lead I ECG, necessitating the removal of such anomalies for accurate analysis. From a clinical perspective, the definition of ECG conditions often requires morphological anomalies in more than one lead. As such, the analysis of lead I can be a strong indicator for specific pathologies but may not provide final evidence if a strictly clinical definition is followed. In line with this and existing literature [54], the pathologies considered in this paper are Right Bundle Branch Block (RBBB), Left Bundle Branch Block (LBBB), Myocardial Infarction (MI), and 1st degree Atrioventricular Block (AV). Regarding MI, only infarction of the left circumflex artery (LCX) was included, given its visibility in the lead I. Generally, however, localization of the infarction area by ECG remains challenging due to individual heart dominance, existing coronary collaterals, and prior tissue damage. Left Atrial Enlargement, Intracardiac Block, and Fascicular Block were excluded from the study. The categorization of sinus rhythm as a healthy sample is due to their general classification as non-pathological.

A k-nearest-neighbor (kNN) classifier was conditioned on the embedded training dataset, consisting of 137924 samples, and its performance was evaluated on a left-out test dataset of 13297 embedded samples from *MedalCare-XL*. The choice of kNN is based on its spatial nature in making predictions, which aligns with the underlying assumption that similar ECGs will be closely positioned in the latent vector space. The use of a simple classifier logic helps to interpret the discriminative power of β -TCVAE. The hyperparameter k of the classifier was determined with the highest accuracy on the *MedalCare-XL* validation dataset. The performance of the classifier was assessed through a confusion matrix. Additionally, explorative data analysis included distribution plots of the encoded data across individual dimensions, marked by different pathologies, to facilitate the identification of distinct patterns and explainability.

Personalization

The evaluation of a personalized model involved fine-tuning the best-performing model for each of 100 randomly selected individuals in the *Icentia11k* dataset. Fig. 4 is a sample embedding from *Icentia11k*

dataset colored by subjects. The scatter plot describes the motivation for personalization as there is a clustering pattern in subjects but not in pathologies due to inter-subject variations, as indicated in related research [19]. Due to the synthetic nature of the *MedalCare-XL* dataset, inter-subject variations may not have been as pronounced and influential in previous anomaly detection. The data embeddings and the corresponding personalized embeddings were split into a training, validation, and test set and proceeded similarly to anomaly detection. The aim was to distinguish normal heartbeats from those that show signs of PVC or PAC. Using t-SNE embeddings, the goodness of separation of these classes was visually investigated. The distributions along the axes of the embedded data were examined for patterns using the interpretations of the dimensions (cf.). Of particular interest was the identification of axes with clear distribution patterns that correspond to the medical facts of the associated pathology.

3. Results

The approach trained 68 models, each targeted for 200 epochs, provided that no early stopping criteria were met. The training included different combinations of parameters (α , β , γ) and latent dimensions (cf. Table 3). The highest MIG of 0.23 was achieved for the parameters $\alpha = 0.2$, $\beta = 0.8$, $\gamma = 0.2$, and a latent vector space dimension of 12. The following results and analyses are derived from this best-performing model.

3.1. Disentanglement

The most significant alignment of features was found in dimensions 3 and 7, which correspond to P-wave and T-wave heights, with R^2 values of 0.82 and 0.89, respectively. The embeddings displayed in Fig. 5 show a clear disentanglement, evidenced by the orthogonal arrangement of the pathways for P- and T-wave heights. Moderate scores and agreement with expert assessments were obtained for S-wave width (dimension 4, $R^2 = 0.3$), P-wave width (dimension 5, $R^2 = 0.24$), QRS/R-Peak width (dimension 6, $R^2 = 0.4$), and QT-interval (dimension 10, $R^2 = 0.14$). The dimension 10 also indicated a characteristic related to the elevation of the ST segment, but was not measured with R^2 as it is not an available factor in the metadata.

Varying values along the dimensions of the embedding space caused an effect on the reconstructions (Fig. 6). The visual examination of the reconstructions revealed only a few altered morphologies per dimension, indicating disentanglement. In this context, the reconstruction plots for the T- and P-wave (dimensions 1 and 7) revealed the most pronounced morphological variations, aligning with the computational analysis. In contrast, features with lower R^2 values, such as the QT-interval, exhibit less distinct morphological changes in the reconstruction. In addition, the phenomenon of a mode collapse occurred in dimensions 0 and 2. Dimensions for which the experts observed a mode collapse (0 and 2) or a high redundancy to another dimension (5) and an unclear morphological change (8) were excluded from the presentation.

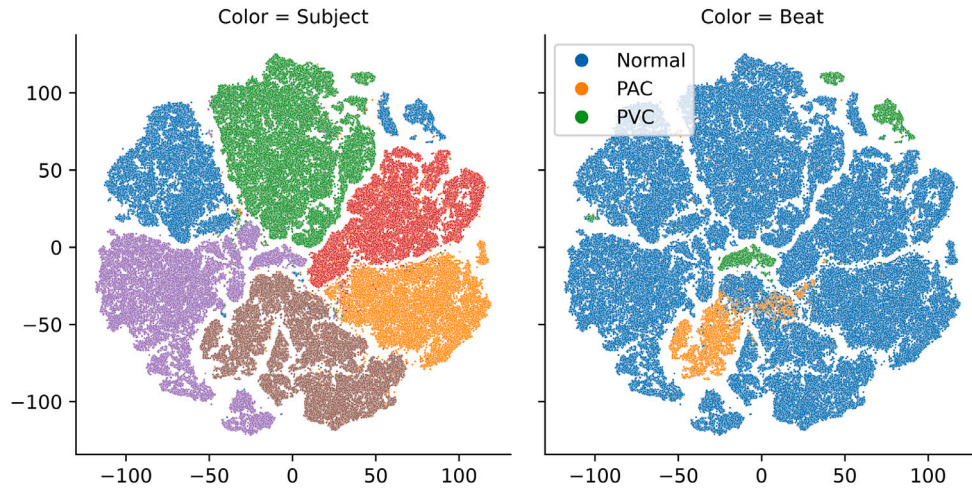


Fig. 4. Common t-SNE embedding of six subjects, colored either by subject (left) or by beat annotation (right).

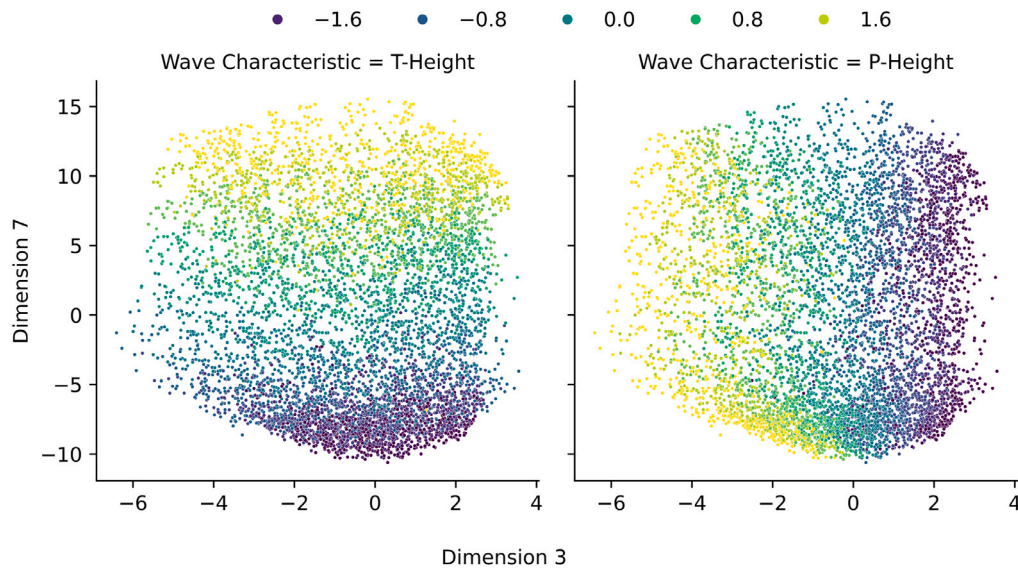


Fig. 5. Slice through dimensions three and seven of the embedding space.

3.2. Anomaly detection

In evaluating the kNN multiclass predictor, the validation set achieved the best accuracy of 82.96% with hyperparameter $k = 39$. When applied to the left-out test set, the classifier demonstrated an overall accuracy of 81.98% and a weighted F1 score of 0.8218. This result reflects the per-class performance as follows: (1) the AVBlock class showed a F1 Score of 0.7582, with 20.09% of samples misidentified as sinus rhythm; (2) the LBBB class achieved 0.7852 F1 score, with 19.54% of its samples incorrectly classified as RBBB; (3) the MI LCX class exhibited a high classification F1 score of 0.9213% (4) the RBBB class recorded a 0.8075% F1 score and mirrored the LBBB class in terms of misclassification, with 16.02% of samples falsely identified as LBBB; (5) the sinus rhythm class, indicative of healthy samples, attained an F1 score of 0.7475%, with 16.62% of predictions misclassified as AVBlock. The provides additional insight into the performance of the multiclass anomaly detection (cf. Fig. 9).

In binary classification, the chosen value of $k = 5$ led to the highest accuracy on the validation set, reaching 92.28%. Analysis of the test set revealed a similar accuracy of 90.49%, supported by a sensitivity of 0.9504, a specificity of 0.7146, and an F1 score of 0.9417.

The investigation of the data distribution along the explainable axes revealed four dimensions with evident split behavior regarding the

pathology classes (cf. Fig. 7). The healthy class sinus rhythm shows a Gaussian distribution around the mean value 0, which is an expected behavior due to the assumption that values near the prior mean are the healthy and most occurring samples.

1. Dimension 3: P-wave height & Q-wave width

Branch blocks, especially RBBB, are distinguishable along dimension 3. This is likely due to the altered Q-shape as part of a generally prolonged QRS complex. A prolonged QRS duration is the key feature of branch blocks, this holds for RBBB as well as for LBBB. The prolonged QRS duration is caused by delayed polarization of the ventricle; as the direct conductive system is blocked, polarization of the blocked ventricle is induced by its unblocked counterpart, leading to an extended total depolarization time. In addition, a reduced Q-wave or even the complete absence thereof in lead I can be a defining characteristic of LBBB. On the contrary, a potential deviation along an axis associated with the height of the P-wave does not seem medically explainable. Neither LBBB nor RBBB are linked to the P-wave. The P-wave reflects atrial depolarization whereas LBBB and RBBB indicate dysfunctional ventricular depolarization. Therefore, deviations related to P-wave height may not be medically explainable in the context of bundle branch blocks.

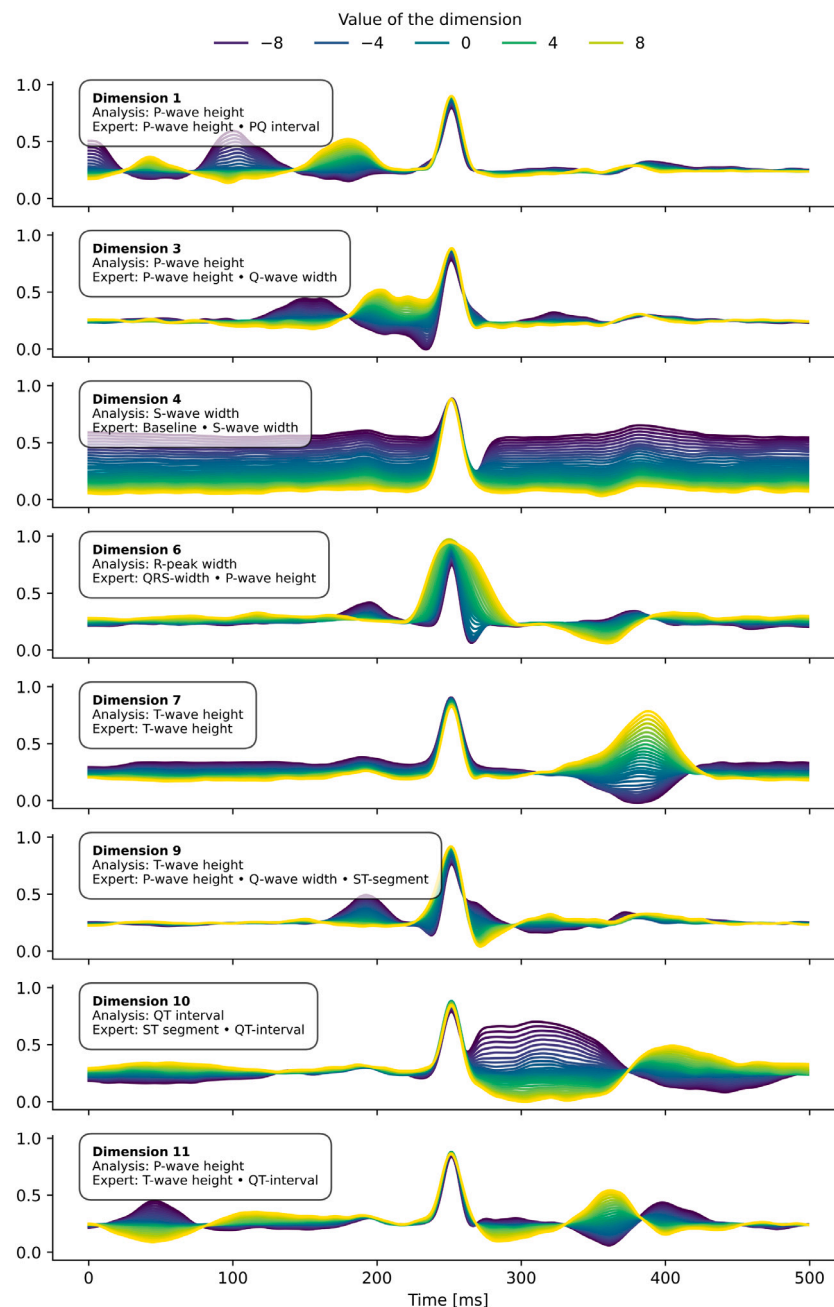


Fig. 6. Reconstructions for varying values along single dimensions and interpretation given by computational analysis and expert.

2. Dimension 4: S-wave width

All classes have a distinct center of mass, indicating a strong separation criterion, although there is overlap, particularly between RBBB and LBBB, as well as AV block and sinus rhythm. Again, a divergence of RBBB and LBBB from 0 can be expected as changes in the S-wave and thus the QRS width constitute a diagnostic feature of bundle branch blocks. More specifically, a slurry S-wave in the first lead is indicative of RBBB in particular. In addition, specific altering ventricular depolarization-configurations (e.g. rSR', a small R-wave followed by a larger S- and another R-wave) may be indicative of RBBB. These alterations, however, usually tend to be seen in other leads than I.

3. Dimension 6: QRS-complex width & P-wave height & R-peak width

A widened QRS-complex serves as an important indicator of bundle branch blocks in general. The QRS-complex width

distinguishes LBBB and most of the RBBB from AV block and sinus rhythm. In addition, this dimension shows a broadened R-peak. Clinically, a broadened R-peak can be indicative for LBBB. This might further explain the separation of LBBB located away from the overall mean of zero.

4. Dimension 10: QT-interval & ST-segment

MI class variation can be seen in the QT- and ST-segment axes. Clinically, variations in the ST-segment range from ST-depression (NSTEMI) to ST-segment elevation (STEMI), both signs of MI. This explains the peak of the MI class on both the left and the right side of normal cases. Some locations of myocardial infarction can typically be seen in lead I. An MI involving the left circumflex artery, for example, affects the lateral wall of the left ventricle and is consequently reflected in lead I. Furthermore, alterations of the Q-wave are also indicative of MI. Depending

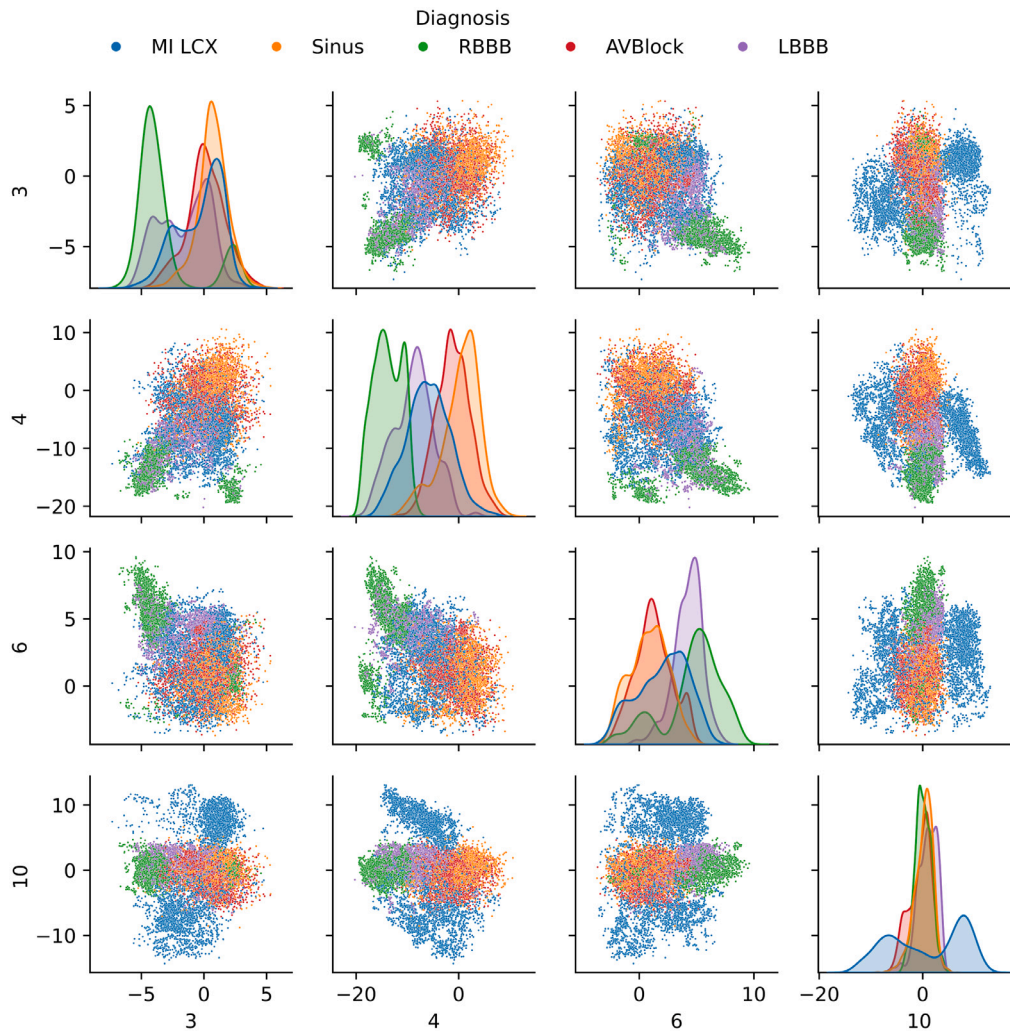


Fig. 7. Distribution of encoded ECG points along their explainable dimensions.

on the phase of the MI, the Q-wave may be pathologically wide and deep.

3.3. Personalization

Fine-tuning the model with a subset of subjects from the *Icentia11k* dataset yielded 100 models, each trained for 20 epochs. The precision for these models (82 out of 100, as some subjects had only one class, making them unsuitable for performance assessment), represented by a confusion matrix summed over all chosen subjects, was 0.9978, 0.8935, and 0.9565 for normal, PAC, and PVC classes, respectively. The equivalent values for recall were 0.9989, 0.7789, and 0.956. The associated per-class based F1 scores were 0.9984, 0.8323, and 0.9563, with an overall macro F1 score of 0.929.

To address potential bias due to the imbalanced class distribution across subjects, this metric was also calculated on a per-subject basis and then averaged. The PAC and PVC classes were merged into a single anomaly class, and only subjects with more than 500 anomaly samples were included. This approach was necessary because some subjects exhibited only PAC, while others had only PVC alongside normal samples. Additionally, a minimum of 100 samples in the test set (20% of 500) was required to prevent strong influence on the average results by subjects with very few samples, where misclassifications, though minimal, could disproportionately affect the overall outcome. 40 subjects remained and were included into the analysis. The average

F1 score was 0.8719, with precisions of 0.9953 and 0.8911, and recalls of 0.9975 and 0.6899 for the normal and anomaly classes, respectively. The lower recall performance for the anomaly class can largely be attributed to the difficulty in distinguishing normal and PAC cases, as PACs can show similar morphology as sinus beats. This is due to the fact that the depolarization initiating cell might be located close to the sinoatrial node. In these cases, differences in P-wave morphology, the timing of the QRS complex, and the presence of compensatory pauses can be subtle [55], especially in short ECG segments without extended rhythm context. In the case of PVCs, however, irregular contractions are initiated within the ventricle apart from the regular conductive systems. This leads to diverging morphologies of the complete QRS-complex and T-wave.

A visual analysis of the t-SNE embeddings, as depicted in Fig. 8(a), for six subjects selected for convenience (with the aim of an even distribution of the classes) indicated pronounced clustering by class, particularly for the PVC samples. The distribution of the embedding along dimension 1, which is associated with the morphology of the P-waves, showed an expected deviation from normality in two cases (cf. Fig. 8(b)). In a third case, there was no deviation in the distribution of the PAC samples, which was the case for PVC. One phenomenon observed with PVC is the absence of a preceding P wave, as PVC occurs independently of sinus node activity.

A deviation in the fourth dimension was expected for PVC cases as it is associated with the S-wave width factor. The strong pattern in Fig. 8(c) for PVC is likely due to the general association of morphological changes in the QRS complex.

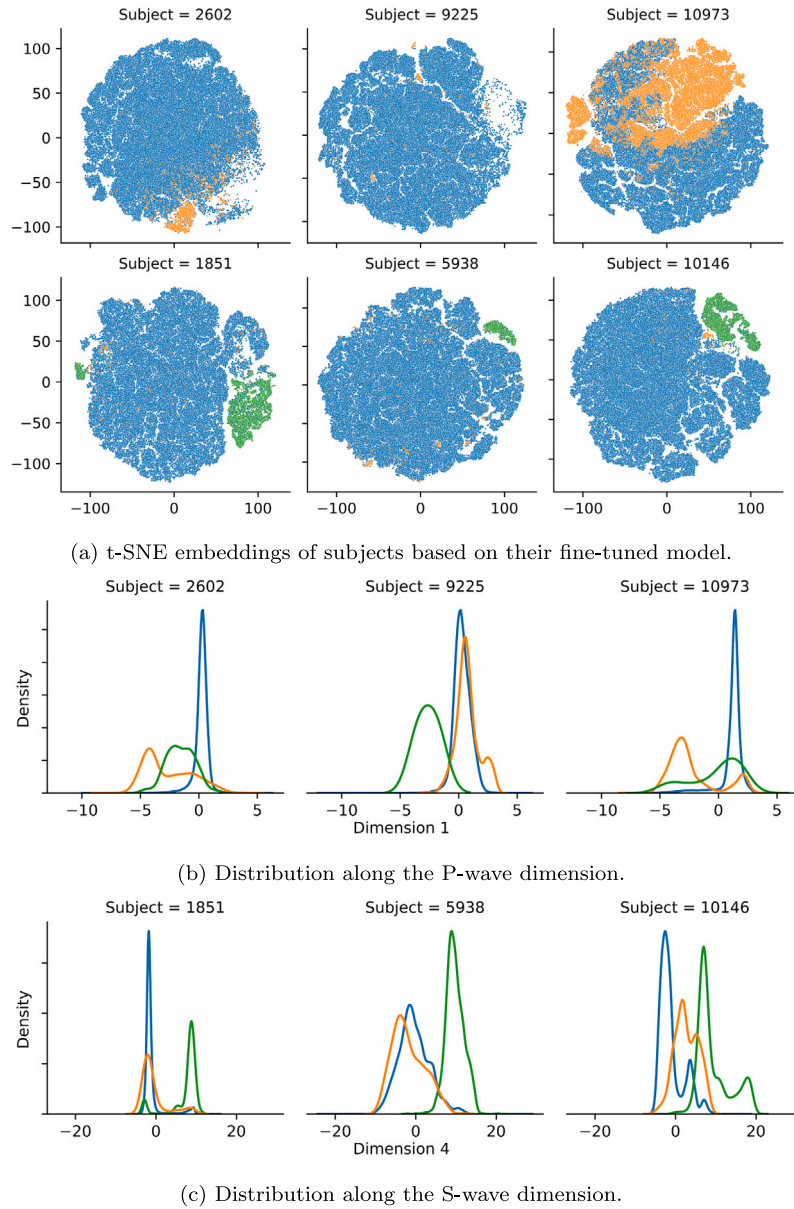


Fig. 8. Embedding space of the Icential1k dataset.

4. Discussion

The results reveal patterns that indicate a structured latent vector space and good model capacities for anomaly detection.

4.1. Disentanglement

A refined disentangled CNN based variational autoencoder was implemented to map morphological ECG factors into single dimensions of the latent vector space to enhance the explainability. By using the MIG as a classifier-free tool for mapping features to dimensions, this research advances the automation of axis interpretation, moving beyond the traditional manual investigation of the reconstruction space, as previously demonstrated in FactorECG [23].

The ECG embeddings along these dimensions revealed a strong to moderate one-to-one alignment with ECG factors, as shown in the embedding and reconstruction plots (cf. Figs. 5 and 6). The results confirm the effectiveness of the method in creating explainable dimensions,

especially in narrowing down to the coarser task of detecting any morphological change of the P-wave, the QRS-complex, and the T-wave. The approach can be particularly relevant for tracking inter-subject variations by embedding vectors that diverge from their respective centers of mass along the explainable axes. However, achieving complete disentanglement of all morphological variations is challenging due to the interdependent nature of the heart's electrical activity and, thus, the morphology of the ECG. For instance, alterations in the S-wave width are linked to modifications in the QRS-complex, and ostensibly independent features like the QT-length and the widths of the Q- and T-wave exhibit practical dependencies. The evaluation, relying on a vast collection of real-world ECGs, may encompass a partial spectrum of potential ECG morphological variations. In contrast, synthetic datasets aim to represent a broad spectrum, but simultaneously altering multiple morphologies faced limitations in generating ECGs. The most accurate ECG simulations were observed in variations of the P-wave and T-wave heights, which are also the features most effectively encoded into single dimensions. This finding supports the model's utility for monitoring

changes related to atrial depolarization (P-wave) and ventricular repolarization (T-wave). The computational interpretation (R^2 scores) of the twelve dimensions roughly matched the experts' assessment by visual examination of the reconstruction plots, although the experts' rating provided much more comprehensive information. A fully automated system that can interpret the dimensions autonomously does not yet seem feasible and still requires the intervention of a trained specialist. In summary, the β -TCVAE provides a viable method to disentangle the representation of ECGs that can detect altered morphologies in an explainable manner.

4.2. Anomaly detection

The kNN classifier, which relies on an intuitive and low-dimensional data structure, performed well in predicting ECG conditions. The predicted values and visual inspection (cf. Fig. 7) indicate strong separability of classes in the latent vector space. The healthy samples (the broad mass of ECGs) are, by default, Gaussian distributed around the prior mean, which is expected behavior. This supports the objective that outliers and anomalies on the dimensions are identifiable using simple techniques such as z-scores. The interpretation of the individual dimensions and the observed deviations for pathologies is in line with medical expertise. For example, dimensions 4 and 6, representing a changing QRS-complex, exhibited a significant deviation in the RBBB and LBBB samples. In addition, MI LCX pathology showed up as a strong dimension 10 pattern associated with the ST segment. A clear distinction, specifically for the classes RBBB and LBBB, as well as sinus rhythm and AV block, is hard to achieve in single beats of lead I ECGs, even for professionals.

MedalCare-XL is a recently published dataset. Few studies have involved this dataset, limiting direct comparisons to the results of this work. In general, the performance in the multiclass scenario falls short of the gold standards with accuracies above 95% [28], which typically relies on multi-lead signals and at the cost of using black-box models. The reported average accuracy, F1 score, sensitivity, precision, and specificity of deep-learning approaches are 90.15%, 0.8389, 0.7816, 0.6282, and 0.9386, respectively, under the more conservative inter-subject evaluation paradigm [28], which was also applied in this study. Notably, the precision for the β -TCVAE reflects an average precision of 0.8011, significantly exceeding the average precision scores of related work. However, comparability is further hindered by variations in anomaly performance assessment across datasets and the predominance of the MIT-BIH dataset in most existing studies. While the accuracy of FactorECG in conventional ECG interpretation is considered high (AUROC of 0.94), 35 binary logistic regression models were created, one for each class, which can be impractical when deploying the approach in resource-constrained environments. The associated black box model in the FactorECG research achieved an AUROC of 0.96 [23].

For the presented binary classification approach, an accuracy of 90.94% and a sensitivity score of 95.04% align with notable heartbeat anomaly detection approaches, which range from 89.38% to 99.77% in accuracy and 86.01% to 99.51% in sensitivity based on the MIT-BIH dataset [56].

4.3. Personalization

The t-SNE embedding of the encoded real-world long-term ECGs from the *Icentia11k* dataset showed pronounced clustering behavior between subjects. Fine-tuning the model to each subject for personalized predictions proved effective in dealing with inter-subject variations. Personalized embedding revealed clustering by pathologies, supported by a high classification performance. In particular, variations in PVC and PAC were aligned to the dimensions closely related to their medical explanations. For example, signs of PVC were identifiable in dimension four, which is linked to S-wave changes (more generally,

QRS-complex), whereas PAC cases associated with the P-wave were detectable in dimension 1.

These results emphasize the potential of an explainable base model that can be adapted to individual ECGs. Especially in resource-constrained and mobile environments, the fine-tuning approach may become feasible to realize the full potential of tracking changes indicative of anomalies. In this context, the challenge of artifacts in the data, induced by the mobile environment, must be considered. These artifacts may have also impacted the performance of this research on the *Icentia11k* dataset, which was similarly recorded in a mobile environment. Despite these challenges, the accuracy scores achieved in this research surpass the balanced accuracy score of 56%, which was the best result of an unsupervised approach reported in the original *Icentia11k* publication based on ten randomly chosen subjects [38].

4.4. Impact and application

The impact and potential applications of this work in cardiology are profound. Continuous tracking of ECG signals using wearable technology enables early detection of cardiac anomalies, which is crucial for timely interventions and reducing the long-term burden on healthcare systems. By focusing on interpretable baseline changes, this approach provides an advantage over traditional systems, offering insights into subtle ECG variations that are medically relevant. The disentangled representational learning enhances the interpretability of outliers along the dimensions of the latent vector space associated with ECG features, allowing for more targeted monitoring of pathologies such as bundle branch blocks and myocardial infarctions. Furthermore, the personalization aspect of the model addresses inter-subject variability, improving diagnostic precision and making it adaptable for individual patients. The ability to scale and integrate into mobile devices extends monitoring beyond the clinical environment, enabling long-term recordings and real-time anomaly detection in everyday settings. Additionally, this system allows for the selection of primarily the detected outliers for further examination, saving time in analyzing large datasets. The synthetic generation of ECGs based on these models further holds promise for training and diagnostic support, enhancing both clinical practice and public health initiatives.

4.5. Limitations

The primary limitation of this study, particularly in anomaly detection, is that it focuses on single beats of lead I ECG signals. Detection of most anomalies requires analysis of multiple leads and consecutive beats, a requirement not met by the current research framework.

Another aspect that needs to be thoroughly explored is the hypothesis that disentanglement may be more influenced by the hyperparameters and architecture of the model than by the originally presented loss target [57]. In line with this, it was observed that not all experiments with increased regularization of the total correlation term resulted in an increased MIG value. This might be due to the inherent interdependencies among various ECG features (e.g., the effect of S-wave width on the QRS complex), which complicate the process of disentangling these signals. Additionally, the presence of mode collapse reduces the amount of pure information mapped to individual dimensions. Consequently, achieving full disentanglement is challenging, as overlapping morphological changes can obscure the clear identification of distinct pathological signals.

This research employed five open-source datasets to account for non-identically and independently distributed data, derived from ECGs collected from various individuals and devices. However, the transferability to other data sources, especially those recorded in unsupervised mobile environments, may lead to reduced performance. This is further evident in the observation that even minor changes in hyperparameters can have a disproportionately large effect on performance scores, raising the limitation and question of hyperparameter validity across different settings.

4.6. Future work

The identified limitations suggest the necessity for expanding the scope beyond a single lead. A proposed direction is to develop a model for each lead, focusing primarily on encoding pure morphological features rather than pathologies. Additionally, the generation of synthetic ECG alterations becomes imperative to comprehensively capture the full spectrum of potential morphological variations. A synthetic dataset that uniformly represents the morphological variations of an ECG would serve as a valuable resource for training a base model, which would become highly explainable based on the information on the generative factors.

The architecture in place addresses the proposed future work of Steenkist et al. (2019) [22] by incorporating non-linear layers. However, even more sophisticated priors and architectures can be explored, such as attention mechanisms, which have shown remarkable results in recent years [58]. Therefore, investigations into various encoder–decoder architectures are necessary, particularly in light of the prevalent challenge of mode collapse observed in the study. While some dimensions demonstrated clear associations with specific features, others appeared to have entangled responsibilities for multiple distinct features, leaving certain dimensions with minimal impact. To address mode collapse and enhance the model’s explanatory power, incorporating more diverse datasets is recommended as a potential solution.

Finally, the motivation for focusing on single-lead ECGs stems from the recording capabilities of wearable devices, which enable the model to be used in mobile environments. Future research directions include fine-tuning the model on mobile devices and incorporating federated learning to continuously improve the base model while preserving privacy.

5. Conclusions

The objective and novelty of this research lies in the application of a refined method for disentangled embedding of multi-source ECG heartbeats, while also addressing the healthcare challenges posed by heterogeneous environments.

Representational learning of single-lead ECG using β -TCVAE showed decent disentanglement properties regarding the morphology of one-second heartbeats. The factors for atrial- and ventricular electrical activity (primarily P- and T-wave) revealed an orthogonal embedding structure, allowing for explainable and reliable outlier detection. Detecting anomalies in the embedding space resulted in high accuracy, especially in the binary classification of healthy and pathological samples. The associated deviations from the healthy samples along the explainable embedding dimensions were predominantly reasonable and supported by medical evidence. Fine-tuning the model to individual ECGs as a method for personalization successfully mitigated inter-subject variations in a real-world, wearable dataset.

The outcome was a structured and accessible latent vector space, where anomaly detection was achieved in an explainable and personalized manner, with a slight trade-off in accuracy compared to non-explainable methods. Specifically, the proposed approach proved effective in binary anomaly detection and showed promise for tracking deviations of embedded vectors along explainable axes.

CRediT authorship contribution statement

Maximilian Kapsecker: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Matthias C. Möller:** Writing – review & editing, Validation. **Stephan M. Jonas:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

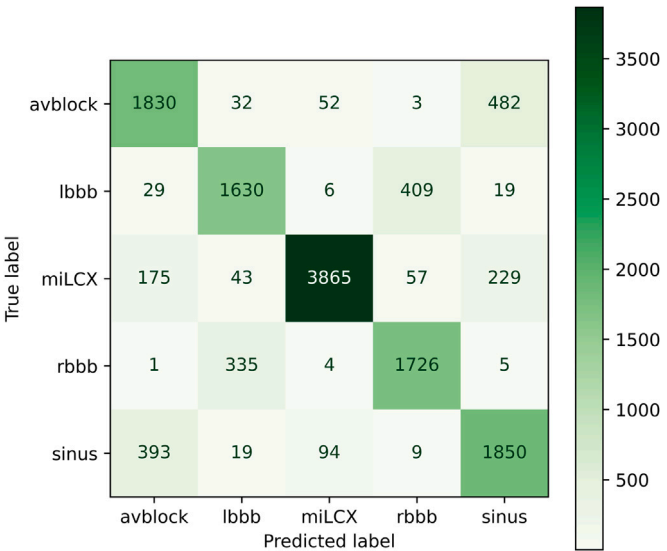


Fig. 9. Confusion matrix for the detection of anomalies, evaluated with the MedalCare-XL test set.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used AI in order to proofread the text and eliminate typos and grammatical flaws. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Systematic model training

In a first run 30 experiments were conducted to find the parameter combination with best MIG. In a refinement round, a further set of 38 models were optimized using parameter-values nearby the best triple (α, β, γ) from the first round. The results are compiled in Table 3.

Anomaly detection performance

See Fig. 9.

Table 3
Systematic variants of the model parameters and associated performance results.

Parameters latent dim	Loss			Disentanglement					
	α	β	γ	Total	MAE	MI	TC	DW KL	MIG (SYNECG)
8	0	2	0.10	50.94	40	5.42	1.70	75.38	0.17
8	0.10	0.40	0.10	48.40	36.85	5.94	4.29	92.44	0.08
8	1.0	1.0	1.0	65.17	49.39	4.04	1.40	10.34	0.12
8	1.0	2	1.0	68.50	55.51	3.70	0.50	8.29	0.06
8	1.0	4	1.0	78.63	56.04	3.41	0.37	17.69	0.12
8	1.0	8	1.0	74.06	56.65	3.14	0.24	12.35	0.05
12	0	2	0.10	53.34	42.43	5.44	1.61	76.95	0.07
12	0.10	0.40	0.10	48.61	35.99	6.10	6.32	94.74	0.08
12	1.0	1.0	1.0	66.47	49.27	4.24	2.43	10.52	0.07
12	1.0	2	1.0	72.78	53.80	3.79	0.54	14.10	0.09
12	1.0	4	1.0	71.25	56.83	3.39	0.33	9.71	0.14
12	1.0	8	1.0	73.54	57.29	3.17	0.20	11.50	0.02
16	0	2	0.10	51.97	41.39	5.44	1.80	69.82	0.12
16	0.10	0.40	0.10	49.62	36.66	6.05	6.73	96.64	0.14
16	1.0	1.0	1.0	67.84	51.65	4.20	2.47	9.51	0.01
16	1.0	2	1.0	71.17	54.19	3.68	0.74	11.81	0.03
16	1.0	4	1.0	75.40	55.16	3.53	0.37	15.24	0.04
16	1.0	8	1.0	70.48	54.44	3.19	0.20	11.24	0.04
20	0	2	0.10	55.19	42.19	5.47	1.89	92.31	0.13
20	0.10	0.40	0.10	52.10	36.72	6.14	7.62	117.14	0.20
20	1.0	1.0	1.0	66.29	49.75	4.16	3.30	9.09	0.05
20	1.0	2	1.0	69.55	53.75	3.85	0.83	10.28	0.04
20	1.0	4	1.0	75.78	55.94	3.48	0.35	14.98	0.05
20	1.0	8	1.0	72.90	57.19	3.22	0.20	10.86	0.02
24	0	2	0.10	53.14	41.31	5.44	1.88	80.63	0.06
24	0.10	0.40	0.10	53.20	43.68	5.99	5.78	66.13	0.16
24	1.0	1.0	1.0	67.12	50.86	4.24	2.93	9.08	0.03
24	1.0	2	1.0	75.21	51.91	4.25	1.11	16.84	0.09
24	1.0	4	1.0	76.28	57.41	3.36	0.27	14.44	0.03
24	1.0	8	1.0	72.55	57.15	3.19	0.16	10.96	0.02
8	0.01	0.04	0.01	34.30	29.17	6.22	10.01	467.54	0.17
8	0.05	0.20	0.05	45.65	35.36	6.05	6.34	174.39	0.19
8	0.10	0.40	0.10	53.05	39.60	6.02	4.69	109.68	0.14
8	0.20	0.80	0.20	54.78	40.78	5.56	2.99	52.48	0.09
8	0.30	1.20	0.30	57.67	44.20	5.28	2.43	29.90	0.17
8	0.40	1.60	0.40	62.68	51.56	4.76	1.20	18.25	0.12
12	0.01	0.04	0.01	34.75	28.67	6.25	16.48	534.93	0.08
12	0.05	0.20	0.05	44.65	32.87	6.19	9.39	191.90	0.12
12	0.10	0.40	0.10	51.93	42.97	5.91	4.36	66.33	0.08
12	0.10	0.40	0.10	48.47	36.01	6.03	5.69	95.82	0.18
12	0.20	0.80	0.20	59.27	42.38	5.85	3.61	64.15	0.23
12	0.30	1.20	0.30	59.48	45.51	5.23	1.63	34.83	0.14
12	0.40	1.60	0.40	62.71	49.18	4.71	1.16	24.48	0.05
12	0.50	2	0.50	64.20	49.39	4.55	0.98	21.15	0.13
12	0.50	2	0.50	64.88	51.83	4.57	0.98	17.58	0.05
16	0.01	0.04	0.01	31.69	24.87	6.25	22.35	587.26	0.05
16	0.05	0.20	0.05	46.93	39.80	6.11	7.72	105.53	0.09
16	0.05	0.20	0.05	41.73	31.02	6.20	10.64	165.39	0.15
16	0.10	0.40	0.10	52.32	42.68	6.04	5.21	69.51	0.05
16	0.20	0.80	0.20	56.09	42.90	5.58	2.77	49.28	0.09
16	0.30	1.20	0.30	64.41	44.46	5.34	2.15	52.59	0.09
16	0.40	1.60	0.40	62.16	46.34	4.92	1.32	29.36	0.06
16	0.50	2	0.50	66.80	54.44	4.31	0.72	17.53	0.07
20	0.01	0.04	0.01	32.52	25.51	6.25	22.88	603.29	0.13
20	0.05	0.20	0.05	42.27	31.45	6.20	9.96	170.23	0.22
20	0.10	0.40	0.10	49.51	37.29	6.05	5.62	93.77	0.09
20	0.20	0.80	0.20	56.01	44.60	5.65	3.01	39.38	0.06
20	0.30	1.20	0.30	58.49	43.69	5.13	1.61	37.75	0.08
20	0.40	1.60	0.40	68.39	47.74	4.96	1.33	41.33	0.09
20	0.50	2	0.50	65.91	53.23	4.43	0.85	17.52	0.11
20	0.50	2	0.50	65.51	48.68	4.51	0.80	25.95	0.06
24	0.01	0.04	0.01	32.67	25.63	6.24	21.66	610.65	0.22
24	0.05	0.20	0.05	42.33	31.26	6.20	10.89	171.61	0.12
24	0.10	0.40	0.10	49.30	36.66	6.09	6.42	94.63	0.06
24	0.20	0.80	0.20	54.11	40.57	5.66	3.01	50.03	0.11
24	0.30	1.20	0.30	60.15	45.28	5.24	1.96	36.51	0.14
24	0.40	1.60	0.40	62.99	47.40	4.90	1.23	29.16	0.06
24	0.50	2	0.50	70.01	48.83	4.54	0.97	33.95	0.08

MAE = Mean absolute error (reconstruction loss); MI = Mutual information; TC = Total correlation; DW KL = Dimensionwise Kullback–Leibler; MIG (SYNECG) = Mutual information gap on synthetic data.

References

- [1] World Heart Federation, World Heart Report 2023, Tech. Rep. 1, World Heart Federation, Geneva, Switzerland, 2023.
- [2] J.P. Ferreira, S. Kraus, S. Mitchell, P. Perel, D. Piñeiro, O. Chioncel, R. Colque, R.A. De Boer, J.E. Gomez-Mesa, H. Grancelli, et al., World Heart Federation Roadmap for Heart Failure, *Global Heart* 14 (3) (2019) 197–214.
- [3] S. Thudumu, P. Branch, J. Jin, J. Singh, A comprehensive survey of anomaly detection techniques for high dimensional big data, *J. Big Data* 7 (2020) 1–30.
- [4] D.P. Kingma, M. Welling, Auto-Encoding Variational Bayes, 2013, arXiv preprint arXiv:1312.6114.
- [5] D. Bank, N. Koenigstein, R. Giryes, Autoencoders, in: *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*, Springer International Publishing, 2023, pp. 353–374.
- [6] P. Li, Y. Pei, J. Li, A comprehensive survey on design and application of autoencoder in deep learning, *Appl. Soft Comput.* 138 (2023) 110176.
- [7] K. Nezamabadi, N. Sardaripour, B. Haghi, M. Forouzanfar, Unsupervised ECG Analysis: A Review, *IEEE Rev. Biomed. Eng.* 16 (2022) 208–224.
- [8] V. Kuznetsov, V. Moskalenko, D. Gribanov, N.Y. Zolotikh, Interpretable Feature Generation in ECG Using a Variational Autoencoder, *Front. Genet.* 12 (2021) 638191.
- [9] J.-H. Jang, T.Y. Kim, H.-S. Lim, D. Yoon, Unsupervised feature learning for electrocardiogram data using the convolutional variational autoencoder, *PLoS One* 16 (12) (2021) e0260612.
- [10] S. Nithya, M.M.S. Rani, Stacked Variational Autoencoder in the Classification of Cardiac Arrhythmia using ECG Signals with 2D-ECG Images, in: 2022 International Conference on Intelligent Innovations in Engineering and Technology, ICIET, IEEE, 2022, pp. 222–226.
- [11] G.B. Moody, R.G. Mark, The impact of the MIT-BIH Arrhythmia Database, *IEEE Eng. Med. Biol. Mag.* 20 (3) (2001) 45–50.
- [12] M. Porumb, S. Stranges, A. Pescapè, L. Pecchia, Precision Medicine and Artificial Intelligence: A Pilot Study on Deep Learning for Hypoglycemic Events Detection based on ECG, *Sci. Rep.* 10 (1) (2020) 170.
- [13] M. Porumb, C. Griffen, J. Hattersley, L. Pecchia, Nocturnal low glucose detection in healthy elderly from one-lead ECG using convolutional denoising autoencoders, *Biomed. Signal Process. Control* 62 (2020) 102054.
- [14] Y. Jiang, P. Zheng, D. Lai, A Semi-supervised Algorithm for Atrial Fibrillation Attack Prediction Using Convolution Auto-encoder of Time Series Signal, in: 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society, EMBC, IEEE, 2023, pp. 1–4.
- [15] S. Nurmaini, A. Darmawahyuni, A.N. Sakti Mukti, M.N. Rachmatullah, F. Firdaus, B. Tutuko, Deep Learning-Based Stacked Denoising and Autoencoder for ECG Heartbeat Classification, *Electronics* 9 (1) (2020) 135.
- [16] R. Gayathiri, et al., Unsupervised Anomaly Detection to handle Imbalanced Datasets using Auto encoders for ECG signal Classification, in: 2023 Third International Conference on Secure Cyber Computing and Communication, ICSCCC, IEEE, 2023, pp. 263–268.
- [17] Y. Li, Z. Chen, D. Zha, M. Du, J. Ni, D. Zhang, H. Chen, X. Hu, Towards Learning Disentangled Representations for Time Series, in: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 3270–3278.
- [18] A.J. Moss, W. Zareba, J. Benhorin, E.H. Locati, W.J. Hall, J.L. Robinson, P.J. Schwartz, J.A. Towbin, G.M. Vincent, M.H. Lehmann, et al., ECG T-wave Patterns in Genetically Distinct Forms of the Hereditary Long QT Syndrome, *Circulation* 92 (10) (1995) 2929–2934.
- [19] P.K. Gyawali, J.V. Murkute, M. Toloubidokhti, X. Jiang, B.M. Horacek, J.L. Sapp, L. Wang, Learning to Disentangle Inter-Subject Anatomical Variations in Electrocardiographic Data, *IEEE Trans. Biomed. Eng.* 69 (2) (2021) 860–870.
- [20] P. Gyawali, Z. Li, C. Knight, S. Ghimire, B.M. Horacek, J. Sapp, L. Wang, Improving Disentangled Representation Learning with the Beta Bernoulli Process, in: 2019 IEEE International Conference on Data Mining (ICDM), IEEE, 2019, pp. 1078–1083.
- [21] P.K. Gyawali, B.M. Horacek, J.L. Sapp, L. Wang, Learning Disentangled Representation from 12-lead Electrograms: Application in Localizing the Origin of Ventricular Tachycardia, in: *Workshops at the Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.
- [22] T. Van Steenkiste, D. Deschrijver, T. Dhaene, Interpretable ECG Beat Embedding using Disentangled Variational Auto-Encoders, in: 2019 IEEE 32nd International Symposium on Computer-Based Medical Systems (CBMS), IEEE, 2019, pp. 373–378.
- [23] R.R. van de Leur, M.N. Bos, K. Taha, A. Sammani, M.W. Yeung, S. van Duijvenboden, P.D. Lambiase, R.J. Hassink, P. van der Harst, P.A. Doevendans, et al., Improving explainability of deep neural network-based electrocardiogram interpretation using variational auto-encoders, *Eur. Heart J.-Dig. Health* 3 (3) (2022) 390–404.
- [24] J.-Y. Franceschi, A. Dieuleveut, M. Jaggi, Unsupervised Scalable Representation Learning for Multivariate Time Series, in: *Advances in Neural Information Processing Systems*, 32, 2019.
- [25] P. Rubel, D. Pani, A. Schloegl, J. Fayn, F. Badilini, P.W. Macfarlane, A. Varri, SCP-ECG V3.0: An enhanced standard communication protocol for computer-assisted electrocardiography, in: 2016 Computing in Cardiology Conference, CinC, IEEE, 2016, pp. 309–312.
- [26] V. van der Valk, D. Atsma, R. Scherptong, M. Staring, Joint Optimization of a β -VAE for ECG Task-Specific Feature Extraction, 2023, arXiv preprint arXiv:2304.06476.
- [27] W. Todo, M. Selmani, B. Laurent, J.-M. Loubes, Counterfactual Explanation for Multivariate Time Series Using A Contrastive Variational Autoencoder, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.
- [28] Q. Xiao, K. Lee, S.A. Mokhtar, I. Ismail, A.L.B.M. Pauzi, Q. Zhang, P.Y. Lim, Deep Learning-Based ECG Arrhythmia Classification: A Systematic Review, *Appl. Sci.* 13 (8) (2023) 4964.
- [29] R.T. Chen, X. Li, R.B. Grosse, D.K. Duvenaud, Isolating Sources of Disentanglement in Variational Autoencoders, in: *Advances in Neural Information Processing Systems*, 31, 2018.
- [30] P. Wagner, N. Strodthoff, R.-D. Boussejot, W. Samek, T. Schaeffter, PTB-XL, a large publicly available electrocardiography dataset (version 1.0.3), 2022, PhysioNet. <https://doi.org/10.13026/kfzx-aw45>.
- [31] J. Zheng, J. Zhang, S. Danioko, H. Yao, H. Guo, C. Rakovski, A 12-lead electrocardiogram database for arrhythmia research covering more than 10,000 patients, *Sci. Data* 7 (1) (2020) 1–8.
- [32] S. Tan, G. Androz, S. Ortiz-Gagné, A. Chamseddine, P. Fecteau, A. Courville, Y. Bengio, J.P. Cohen, Icentia11k Single Lead Continuous Raw Electrocardiogram Dataset (version 1.0), 2022, PhysioNet. <https://doi.org/10.13026/kk0v-r952>.
- [33] K. Gillette, M.A. Gsell, C. Nagel, J. Bender, B. Winkler, S.E. Williams, M. Bär, T. Schäffter, O. Dössel, G. Plank, et al., MedaCare-XL: 16,900 healthy and pathological synthetic 12 lead ECGs from electrophysiological simulations, *Sci. Data* 10 (1) (2023) 531.
- [34] P.E. McSharry, G.D. Clifford, L. Tarassenko, L.A. Smith, A dynamical model for generating synthetic electrocardiogram signals, *IEEE Trans. Biomed. Eng.* 50 (3) (2003) 289–294.
- [35] P. Wagner, N. Strodthoff, R.-D. Boussejot, D. Kreisler, F.I. Lunze, W. Samek, T. Schaeffter, PTB-XL, a large publicly available electrocardiography dataset, *Sci. Data* 7 (1) (2020) 1–15.
- [36] A.L. Goldberger, L.A. Amaral, L. Glass, J.M. Hausdorff, P.C. Ivanov, R.G. Mark, J.E. Mietus, G.B. Moody, C.-K. Peng, H.E. Stanley, PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals, *Circulation* 101 (23) (2000) e215–e220.
- [37] N. Saki, M. Karandish, B. Cheraghian, H. Heybar, S.J. Hashemi, M. Azhdari, Prevalence of cardiovascular diseases and associated factors among adults from southwest Iran: Baseline data from hoveyeh cohort study, *BMC Cardiovasc. Disorders* 22 (1) (2022) 1–10.
- [38] S. Tan, G. Androz, S. Ortiz-Gagné, A. Chamseddine, P. Fecteau, A. Courville, Y. Bengio, J.P. Cohen, Icentia11k: An Unsupervised Representation Learning Dataset for Arrhythmia Subtype Discovery, in: *Computing in Cardiology Conference (CinC)*, 2021.
- [39] D. Makowski, T. Pham, Z.J. Lau, J.C. Brammer, F. Lespinasse, H. Pham, C. Schölzel, S.H.A. Chen, NeuroKit2: A Python toolbox for neurophysiological signal processing, *Behav. Res. Methods* 53 (4) (2021) 1689–1696, <http://dx.doi.org/10.3758/s13428-020-01516-y>.
- [40] TensorFlow Datasets, A collection of ready-to-use datasets, <https://www.tensorflow.org/datasets>.
- [41] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, et al., TensorFlow: A System for Large-Scale Machine Learning, in: 12th USENIX Symposium on Operating Systems Design and Implementation, OSDI 16, 2016, pp. 265–283.
- [42] W.B. Johnson, J. Lindenstrauss, G. Schechtman, Extensions of lipschitz maps into Banach spaces, *Israel J. Math.* 54 (2) (1986) 129–138.
- [43] I.T. Jolliffe, J. Cadima, Principal component analysis: a review and recent developments, *Philos. Trans. Royal Soc.a: Math. Phys. Eng. Sci.* 374 (2065) (2016) 20150202.
- [44] L. Van der Maaten, G. Hinton, Visualizing Data using t-SNE, *J. Mach. Learn. Res.* 9 (11) (2008).
- [45] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, A. Lerchner, β -VAE: Learning Basic Visual Concepts with a Constrained Variational Framework, in: *International Conference on Learning Representations*, 2016.
- [46] B. Esmaili, H. Wu, S. Jain, A. Bozkurt, N. Siddharth, B. Paige, D.H. Brooks, J. Dy, J.-W. Meent, Structured Disentangled Representations, in: *The 22nd International Conference on Artificial Intelligence and Statistics*, PMLR, 2019, pp. 2525–2534.
- [47] Z. Ebrahimi, M. Loni, M. Daneshdatab, A. Gharehbaghi, A review on deep learning methods for ECG arrhythmia classification, *Expert Syst. Appl.* X 7 (2020) 100033.
- [48] L. Zhao, Z. Zhang, A improved pooling method for convolutional neural networks, *Sci. Rep.* 14 (1) (2024) 1589.
- [49] D. Yang, K.M. Ngoc, I. Shin, M. Hwang, DPRELU: Dynamic Parametric Rectified Linear Unit and Its Proper Weight Initialization Method, *Int. J. Comput. Intell. Syst.* 16 (1) (2023) 11.

- [50] A.L. Maas, A.Y. Hannun, A.Y. Ng, et al., Rectifier Nonlinearities Improve Neural Network Acoustic Models, in: Proc. Icml, Vol. 30, No. 1, Atlanta, GA, 2013, p. 3.
- [51] J. Lucas, G. Tucker, R. Grosse, M. Norouzi, Understanding Posterior Collapse in Generative Latent Variable Models, 2019, URL <https://openreview.net/forum?id=r1xaVLUYuE>.
- [52] J. He, D. Spokoyny, G. Neubig, T. Berg-Kirkpatrick, Lagging Inference Networks and Posterior Collapse in Variational Autoencoders, 2019, arXiv preprint [arXiv: 1901.05534](https://arxiv.org/abs/1901.05534).
- [53] G. Hinton, Mini-batch Gradient Descent, 2012, Coursera - Neural Networks for Machine Learning, Lecture 6e. (Accessed: 16 February 2024).
- [54] M.P. Witvliet, E.P. Karregat, J.C. Himmelreich, J.S. de Jong, W.A. Lucassen, R.E. Harskamp, Usefulness, pitfalls and interpretation of handheld single-lead electrocardiograms, *J. Electrocardiol.* 66 (2021) 33–37.
- [55] M. Orini, S. van Duijvenboden, W.J. Young, J. Ramírez, A.R. Jones, A. Tinker, P.B. Munroe, P.D. Lambiase, Premature atrial and ventricular contractions detected on wearable-format electrocardiograms and prediction of cardiovascular events, *Eur. Heart J.-Dig. Health* 4 (2) (2023) 112–118.
- [56] H. Li, P. Boulanger, A Survey of Heart Anomaly Detection Using Ambulatory Electrocardiogram (ECG), *Sensors* 20 (5) (2020) 1461.
- [57] F. Locatello, S. Bauer, M. Lucic, G. Raetsch, S. Gelly, B. Schölkopf, O. Bachem, Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations, in: International Conference on Machine Learning, PMLR, 2019, pp. 4114–4124.
- [58] D. Soydaner, Attention mechanism in neural networks: where it comes and where it goes, *Neural Comput. Appl.* 34 (16) (2022) 13371–13385.